

Advait Deshmukh

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Boulder, CO, USA

EDUCATION

- **Doctor of Philosophy, Computer Science** Jan 2025 – Dec 2030
University of Colorado Boulder
Advised by Dr. Maria Antoniak Boulder, CO
- **Master of Science, Computer Science** Aug 2023 – Dec 2025
University of Colorado Boulder Boulder, CO
GPA: 4.0/4.0
- **Bachelor of Engineering, Computer Science** Aug 2017 – Aug 2021
Birla Institute of Technology and Science, Pilani Pilani, India

PUBLICATIONS

C=CONFERENCE, S=IN SUBMISSION,

- [C.1] Advait Deshmukh, Nora Benedict, Melanie Walsh, and Maria Antoniak. 2026. The Garden of Forking Prompts: How Users Explore Narrative Space in Story Generation. In Proceedings of the Conference on Language Modeling (COLM), Oct 2026, <https://colm.eventhosts.cc/Conferences/2026/AcceptedPapers>.
- [C.2] Rohan Das, Advait Deshmukh, Alexandria Leto, Zohar Naaman, I-Ta Lee, and Maria Leonor Pacheco. 2026. A Structured Clustering Approach for Inducing Media Narratives. Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics, <https://aclanthology.org/2026.acl-long.1970/>.
- [C.1] Advait Deshmukh, Ashwin Umadi, Dananjay Srinivas, and Maria Leonor Pacheco. 2025. All entities are not created equal: Examining the long tail for fine-grained entity typing. Proceedings of the 14th Joint Conference on Lexical and Computational Semantics 2025, <https://aclanthology.org/2025.starsem-1.15>.

SELECTED RESEARCH PROJECTS

- **A Structured Clustering Approach for Inducing Media Narratives** Spring 2025 – Summer 2025
Collaborator BLAST Lab, CU Boulder
 - Implemented a modified DAPrompt (Xiang et al.) based relationship classifier to extract causally related event pairs across 4 datasets.
 - Infused knowledge from teacher models (Llama3, Qwen3) to aid in domain adaptation.
- **Examining the Long Tail for Fine-Grained Entity Typing** Fall 2024 – Spring 2025
Independent study advised by Dr. Maria Pacheco BLAST Lab, CU Boulder
 - Conducted rigorous empirical analysis to study limitations of predominantly popular language model-based approaches for ultra-fine entity typing.
 - Developed proxies to predict entity frequencies based on pretrained language models and analyzed its effect on typing performance.
- **Improving LENS with Knowledge Distillation** Spring 2024
Independent study advised by Dr. Tom Yeh BAIC Lab, CU Boulder
 - Devised a selective caption filtering module on top of the approach provided by LENS (Berrios et al.) which enabled text only Large Language Models (LLMs) to "see".
 - Leveraged LLMs and knowledge distillation techniques to train a compact RoBERTa-based classifier model to streamline the filtering process.
- **RealTalk: Real-Time Talk Move Analysis for Online Classrooms** Spring 2024
Course Project CSCI 5922, CU Boulder
 - Developed approaches in real-time talkmove analysis to aid online classrooms especially in low-resource settings.
 - Explored quantization strategies for efficient inference while maintaining near-SOTA performance on the task.
- **Black-Box Machine-Generated Text Detection** Fall 2023
Course Project CSCI 5832, CU Boulder
 - Developed an ensemble system for SemEval 2023 Task 8 that outperformed the task baseline.
 - The ensemble consisting of RoBERTa, BERT, XLNet, and ELECTRA, achieved an f1 score of 87.6 on dev set.

TEACHING EXPERIENCE

- **University of Colorado Boulder** Fall 2024 – Spring 2025
Boulder, CO
Teaching Assistant, CSCI 1300: Intro to Computing
 - Assisted in the course by grading, holding recitations & office hours to support students.
 - Collaborated with faculty to redesign course materials, improving student engagement.
- **University of Colorado Boulder** Fall 2023 – Spring 2024
Boulder, CO
Course Manager, CSCI 1300: Intro to Computing
 - Managed operations for the course, serving over 900 students through office hours and online platform Ed.
 - Collaborated with faculty to create and grade assignments, projects & exams.

PROFESSIONAL EXPERIENCE

- **Societe Generale Global Solution Centre** Jul 2021 – Jul 2023
Bangalore, India
Software Development Engineer
 - Developed a Retrieval Augmented Generation (RAG) based chat system to assist support teams in efficiently handling policy queries.
 - Built an innovation portal, generating 120 ideas across 7 departments to foster cross-departmental collaboration.
- **Centre for Development of Advanced Computing (CDAC)** Jul 2020 – Dec 2020
Pune, India
Research Intern
 - Engineered a scalable platform for High Performance Computing (HPC) job submission to optimize data processing.
 - Developed a portal for seamless management of big data tasks, enhancing research efficiency.
- **VoiceQube** May 2019 – Jul 2019
Bangalore, India
Summer Intern
 - Participated in summer internship program as part of the practice school program at BITS Pilani.
 - Developed a conversational service to assist students in their GRE preparation.

SKILLS

- **Programming Languages:** Python, JavaScript, TypeScript, C++, Java
- **Frameworks & Tools:** React, Spring, FastAPI, Flask
- **Machine Learning Libraries:** PyTorch, Transformers, NLTK, Pandas, Numpy, Langchain, Ollama
- **Cloud Technologies:** Google Cloud, AWS, Microsoft Azure
- **DevOps & Version Control:** Docker, Git, Kubernetes

LANGUAGE SKILLS

- **English:** Proficient

AWARDS AND GRANTS

- **Conference Support funds** – \$1500 Dec 2025
Boulder, USA
Computer Science Department, University of Colorado
- **Travel Subsidy Recipient** – \$750 Nov 2025
Suzhou, China
StarSEM 2025, Co-located with EMNLP 2025
- **Lloyd Botway Fellow** – \$500 April 2025
Boulder, USA
Computer Science Department, University of Colorado
- **Outstanding Teaching Assistant** – \$500 March 2025
Boulder, USA
CSCI 1300, University of Colorado
- **Winner - Poster Competition** – \$500 Feb 2025
Boulder, USA
Annual Research Expo, University of Colorado
- **Grad App Aid Recipient** – \$500 Nov 2024
Boulder, USA
QueerInAI

LEADERSHIP EXPERIENCE

- **Founder and President** Sep 2019 – May 2021
VoiceQube Pilani
 - Founded and led VoiceQube BITS Pilani Chapter to develop Voice interface-based projects.
 - Deployed BITS Buddy to handle queries about campus life, responding to ~4000 requests between January and March 2020.